PROJECT AND TEAM INFORMATION

Personalize Learning Path Optimization

The system turns courses and topics into a living learning map, where each prerequisite guides the next step forward. It adapts to each student by taking into account what they already know, how challenging they want their journey to be, and their personal learning preferences.

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

The problem we want to solve is that every student learns in a different way and at a different speed. Most learning systems provide courses in a fixed order, which may not be suitable for every student. Some students may already know certain topics, while others may need more time to understand them. Also, some advanced topics cannot be studied without first learning their basic topics.
Our project, Personalized Learning Path Optimization, aims to solve this problem by creating a learning path according to the needs of each student. In this system, courses and topics will be represented as nodes, and their prerequisite relationships will be represented as directed edges. The system will check the student's existing knowledge, completed topics, difficulty level of courses, and available study time. Based on these factors, it will suggest a suitable order in which the student should study the topics.
This project is important because it can help students save time and avoid studying topics that they already know. It can also prevent students from starting difficult topics before learning the required basics. The project will use data structures such as graphs, queues, and sorting in C, along with object-oriented programming concepts in C++. This makes the project useful for both learning and applying programming concepts to solve a real-life problem.

State of the Art / Current solution

At present, most students plan their learning using school or college timetables, course syllabuses, online learning platforms, or advice from teachers. These methods usually provide a fixed order of topics that is the same for all students. Students can also make their own study plans based on their interests, previous knowledge, and available time.
Some online learning platforms recommend courses based on a student's progress or completed lessons. However, these recommendations may not always consider all factors together, such as the student's current knowledge, course difficulty, prerequisite topics, and available study time. Teachers can provide personal guidance, but it may be difficult for them to create a separate learning plan for every student, especially when there are many students in a class.
Therefore, the current methods are useful but may not provide a fully personalized and automatic learning path. Our project aims to improve this by considering different student requirements and automatically arranging courses in a suitable order while making sure that all important prerequisites are completed first.
Existing learning platforms can recommend courses based on student progress. Research systems also use knowledge graphs to represent relationships between learning topics. However, many approaches focus on recommendation based on learner performance and may require more complex machine learning techniques. Our proposed system uses a simpler graph based approach that combines prerequisite relationships, student knowledge, difficulty and available study time.

Project Goals and Milestones

The main goal of our project is to make a system that can help students decide what they should study and in what order. Since every student has different knowledge, difficulty levels, and available study time, the same learning plan may not work for everyone. Our system will try to give each student a suitable learning path. We also want to use the concepts we have learned in Data Structures in C and Object-Oriented Programming in C++. We will use a graph to show the relationship between different courses and their prerequisites. We will also use C++ classes and objects to store and manage information about students and courses.

The initial milestones of our project are:

- *Understanding the problem and planning the project.*
- *Collecting information about courses, topics, and their prerequisites.*
- *Designing the graph to represent the courses.*
- *Storing student information such as their knowledge and available study time.*
- *Implementing prerequisite checking and arranging the courses in the correct order.*

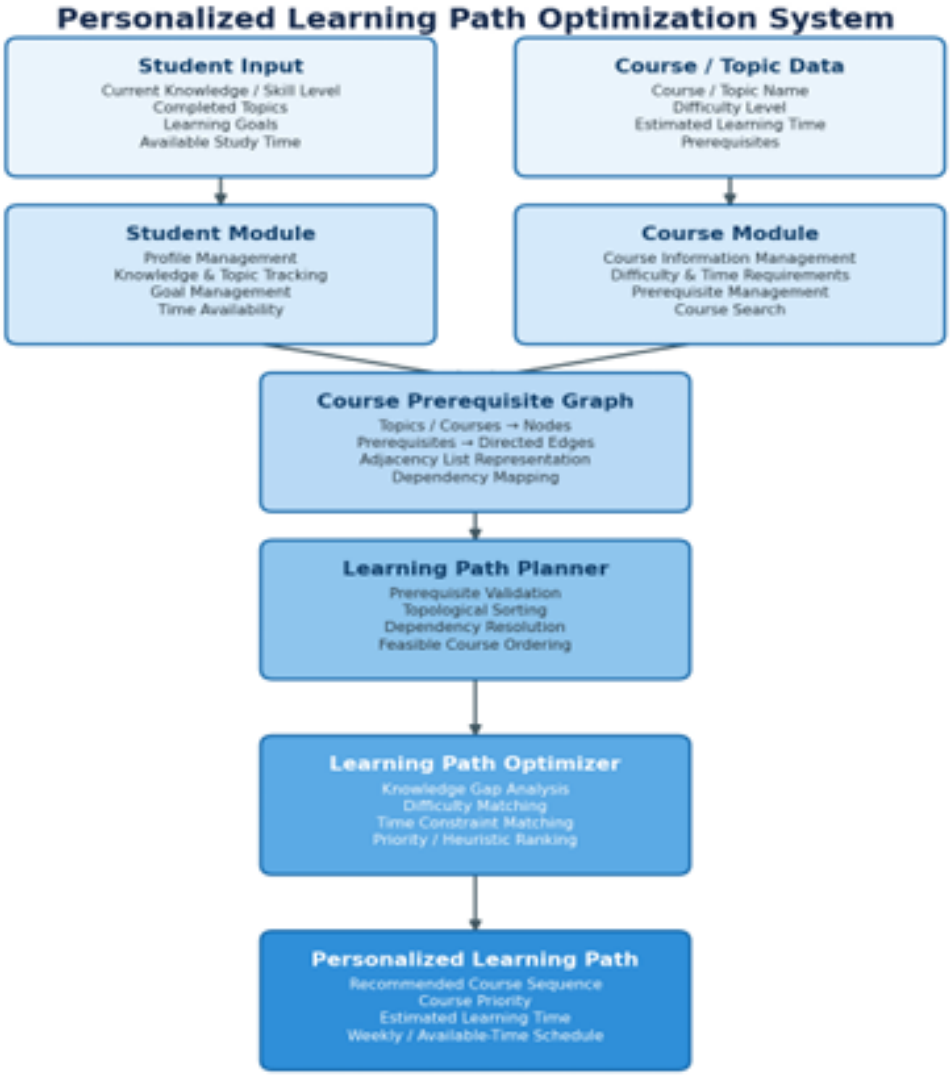
The next milestones will include:

- *Combine the C data structure concepts with the C++ application.*
- *Add the learning time and difficulty-based selection.*
- *Generate and display the final personalized learning path.*
- *Test the system using different student profiles and course combinations.*
- *Find and fix errors in the program.*
- *Improve the system based on testing results.*
- *Prepare the final report, presentation, and project demonstration.*

Project Approach

We plan to solve this problem by creating a simple system that takes information about a student and the available courses and then gives a suitable learning path. First, we will collect details about the courses, such as their name, difficulty level, time required, and prerequisites. We will represent the courses as nodes of a directed graph and the prerequisite relationships as edges. This will help us understand which course should be completed before another course. For the Data Structures part, we will use C to implement concepts such as graphs, arrays, structures, queues, and sorting. We will also use topological sorting to arrange the courses in an order where the required prerequisites are completed first. For the Object-Oriented Programming part, we will use C++. We will create classes such as Student, Course, LearningPath, and Optimizer to store and manage the information properly. Concepts like classes, objects, encapsulation, and functions will be used in the project. The project will mainly be developed and tested on a computer using a C/C++ compiler. We can use platforms such as Visual Studio Code or Dev-C++ for writing and running the program. The project will initially be a console-based application so that we can focus on the main logic and concepts. After developing the basic system, we will test it with different student details and course combinations. We will check whether the generated learning path follows all prerequisites and fits the student's available time. Based on the results, we will correct errors and improve the system. In the final stage, we will prepare the project report and demonstrate how the system works.

System Architecture



Project Outcome / Deliverables

The main outcome of this project will be a Personalized Learning Path Optimization System that helps students choose the right order of courses or topics based on their current knowledge, difficulty level, course requirements, and available study time. The system will represent courses or topics as nodes in a graph and their prerequisite relationships as directed edges. It will use data structures and algorithms such as graphs, adjacency lists, and topological sorting to create a valid learning sequence. The project will also use object oriented programming in C++ to manage student profiles, course information, learning goals, and time availability. The system will analyse the knowledge gaps of a student and give priority to suitable courses. The final deliverables will include the working software, source code in C and C++, system design diagram, algorithms used, sample input and output, test results, and a project report. The final system will provide a recommended learning sequence along with course priority, estimated learning time, and a suitable study schedule.

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